

Introduction to Orthogonal Frequency Division Multiplexing (OFDM) Technique

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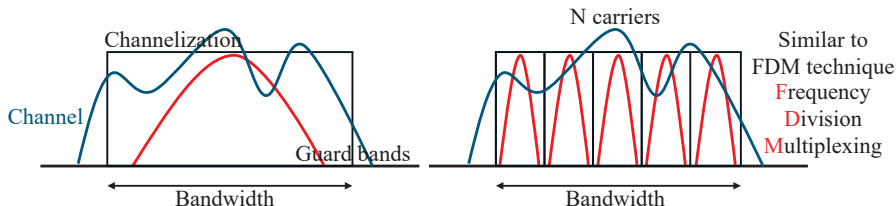
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Outline

- OFDM Overview
- OFDM System Model
- Orthogonality
- Multi-Carrier Equivalent Implementation by Using IDFT (IFFT)
- Cyclic Prefix (CP)
- Summary

Transmission Mode

- Single-Carrier (SC) vs. Multi-Carrier (MC)



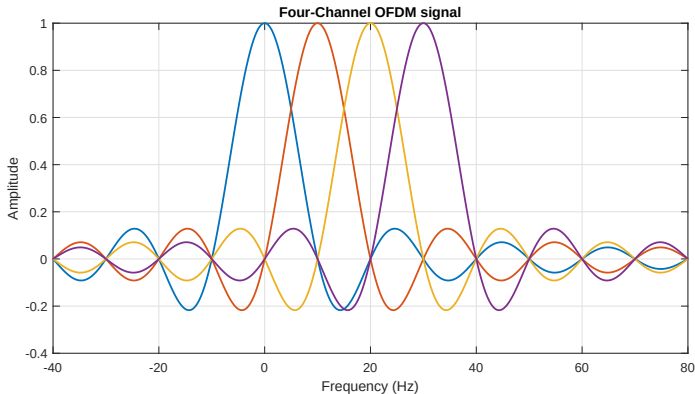
- Data are transmitted over only one carrier.
- Frequency selective fading.
- Data are shared among several carriers and simultaneously transmitted.
- Flat fading.

OFDM Definition

- OFDM
 - **O**rthogonal **F**requency **D**ivision **M**ultiplexing
- The basic principle of OFDM is to **split a high-rate data stream into a number of lower rate streams** that are transmitted simultaneously over a number of sub-carriers.
- It eliminates or alleviates the problem of **multi-path channel fading effect, low spectrum efficiency, and frequency selective fading.**

OFDM Overview(1/2)

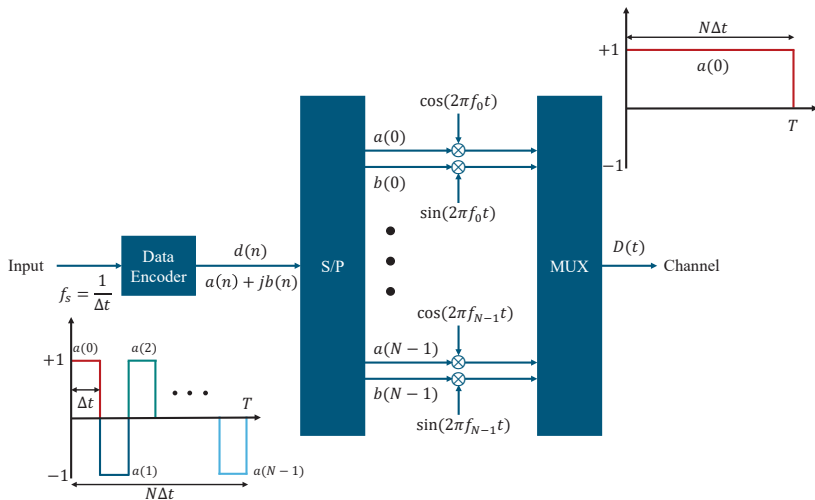
- OFDM modulation



OFDM Overview(2/2)

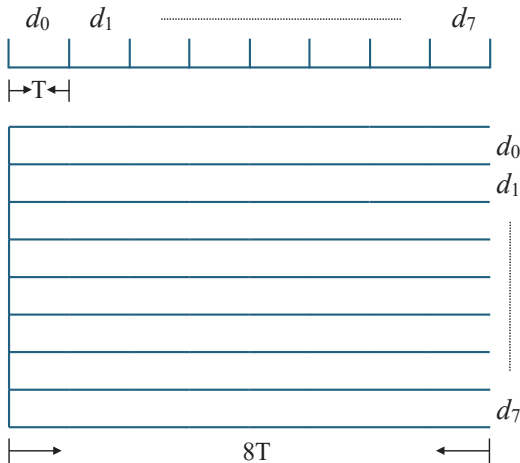
- Feature
 - No intercarrier guard bands
 - Overlapping of bands
 - Spectral efficiency
 - Easy implementation by IFFTs
 - Very sensitive to synchronization

OFDM Transmitter



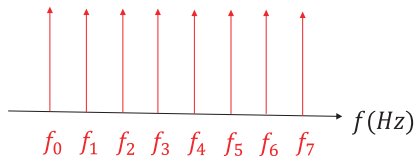
Multi-carrier Systems

- Multi-carrier Transmission



Multi-carrier Systems

- Multi-carrier for $N = 8$



$$\delta(f - f_0) \longrightarrow \exp(j2\pi f_0 t)$$

$$\delta(f - f_1) \longrightarrow \exp(j2\pi f_1 t)$$

$$\delta(f - f_2) \longrightarrow \exp(j2\pi f_2 t)$$

$$\delta(f - f_3) \longrightarrow \exp(j2\pi f_3 t)$$

$$\delta(f - f_4) \longrightarrow \exp(j2\pi f_4 t)$$

$$\delta(f - f_5) \longrightarrow \exp(j2\pi f_5 t)$$

$$\delta(f - f_6) \longrightarrow \exp(j2\pi f_6 t)$$

$$\delta(f - f_7) \longrightarrow \exp(j2\pi f_7 t)$$

Transmitted Signal (1/4)

- The transmitted waveform $D(t)$ can be proved as

$$\begin{aligned} D(t) &= \operatorname{Re} \left\{ \sum_{n=0}^{N-1} d(n) \exp(j2\pi f_n t) \right\} \\ &= \operatorname{Re} \left\{ \sum_{n=0}^{N-1} [a(n) + jb(n)] [\cos(2\pi f_n t) + j \sin(2\pi f_n t)] \right\} \\ &= \sum_{n=0}^{N-1} a(n) \cos(2\pi f_n t) - b(n) \sin(2\pi f_n t) \end{aligned}$$

Transmitted Signal (2/4)

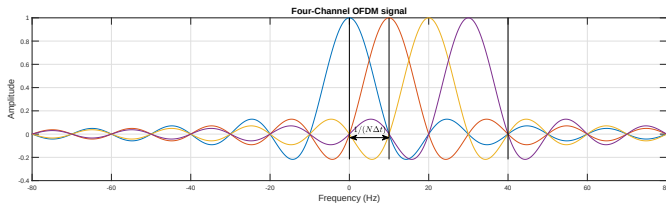
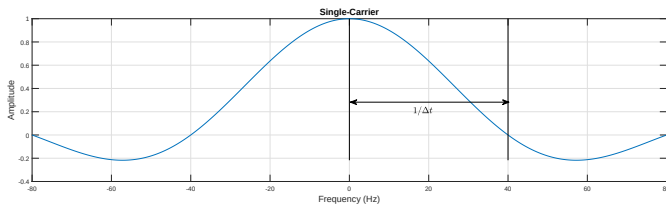
- The transmitted waveform $D(t)$ can be expressed as

$$D(t) = \sum_{n=0}^{N-1} \{a(n) \cos(2\pi f_n t) + b(n) \sin(2\pi f_n t)\} \quad (1)$$

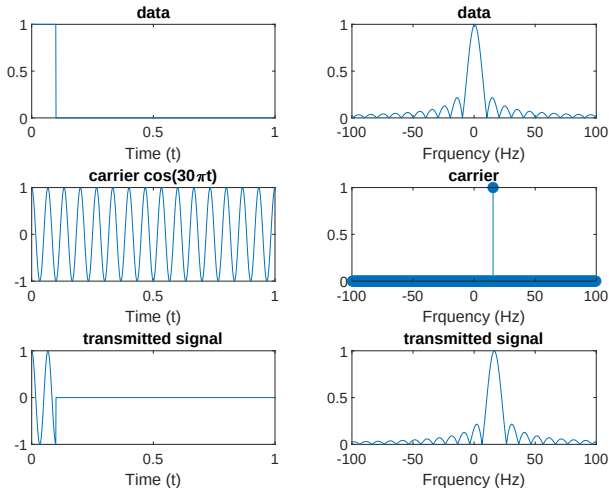
$$\text{where } f_n = f_0 + n\Delta f \text{ and } \Delta f = \frac{1}{N\Delta t}$$

- Using a two-dimensional digital modulation format, the data symbols $d(n)$ can be represented as $a(n) + jb(n)$.
 - $a(n)$: in-phase component
 - $b(n)$: quadrature component

Transmitted Signal (3/4)

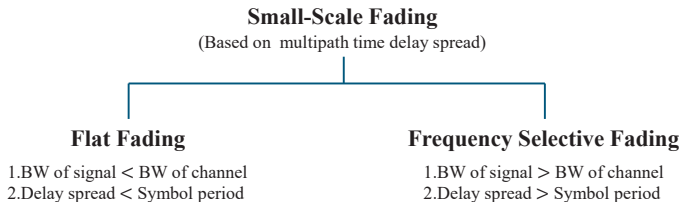


Transmitted Signal (4/4)



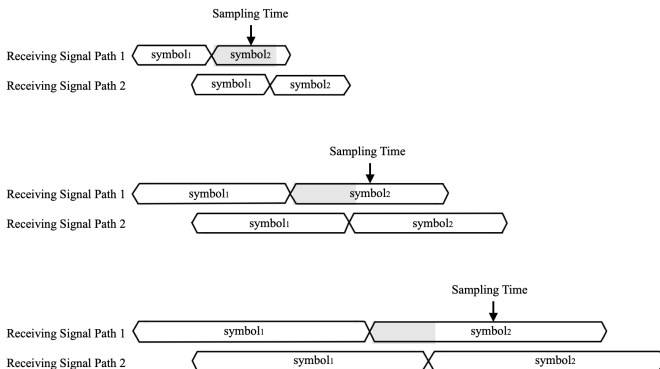
OFDM System Model

- The serial data elements spaced by Δt are grouped and used to modulate N carriers. Thus they are frequency division multiplexed.
- The signaling interval is then increased to $N\Delta t$, which makes the system less susceptible to channel delay spread impairments.



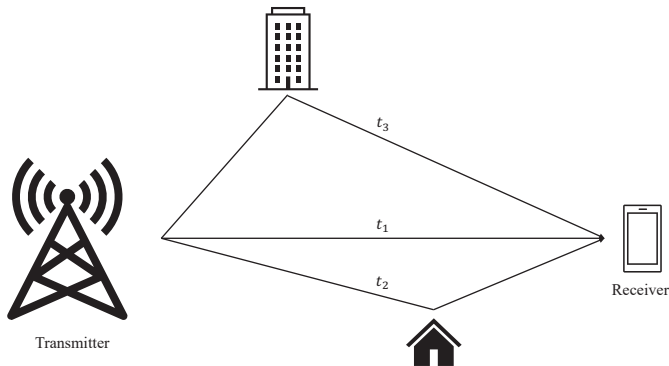
Symbol Period vs. Multipath Interference

- Under the same multipath delay condition, as the symbol period lengthens, the impact of multipath interference decreases.



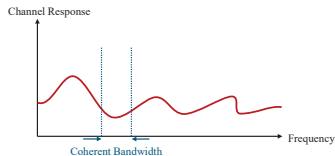
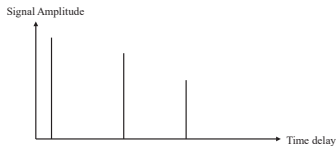
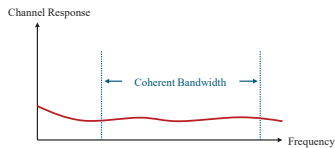
Multipath propagation (1/6)

- Multipath is the propagation phenomenon that results in radio signals reaching the receiving antenna by two or more paths.



Multipath propagation (2/6)

- Multipath propagation results in a received signal that is dispersed in time.



Multipath propagation (3/6) — Proof

- Two-path channel impulse response:

$$h(t) = a\delta(t) + b\delta(t - \tau)$$

where a and b are the amplitudes, and τ is the delay of the second path.

- Frequency response:

$$\begin{aligned} H(f) &= a + be^{-j2\pi f\tau} \\ |H(f)|^2 &= H(f)H(f)^* \\ &= (a + be^{-j2\pi f\tau})(a + be^{j2\pi f\tau}) \\ &= a^2 + ab(e^{-j2\pi f\tau} + e^{j2\pi f\tau}) + b^2 \\ &= a^2 + ab \cdot 2\cos(2\pi f\tau) + b^2 \\ &= a^2 + b^2 + 2ab\cos(2\pi f\tau) \end{aligned}$$

Multipath propagation (4/6) — Proof

- $a = b$.

$$\begin{aligned}|H(f)|^2 &= a^2 + a^2 + 2a^2 \cos(2\pi f\tau) \\ &= 2a^2 (1 + \cos(2\pi f\tau))\end{aligned}$$

If $\cos(2\pi f\tau) = -1$, then signal cancellation.

- $a \neq b$.

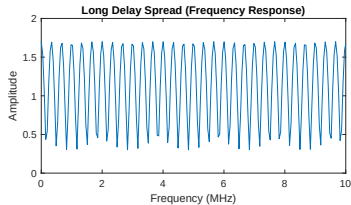
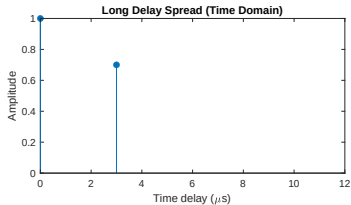
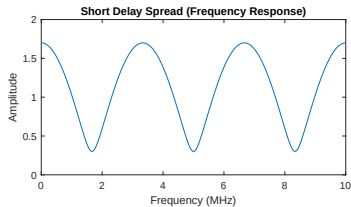
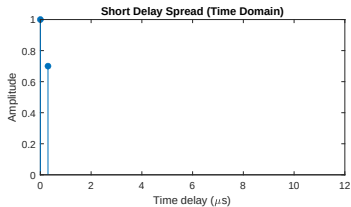
$$|H(f)|^2 = a^2 + b^2 + 2ab \cos(2\pi f\tau)$$

Although $\cos(2\pi f\tau) = -1$, signal is not cancellation.

$$\begin{aligned}|H(f)|_{\min}^2 &= a^2 + b^2 - 2ab = (a - b)^2 \\ |H_{\min}| &= |a - b| \neq 0\end{aligned}$$

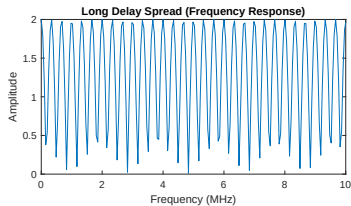
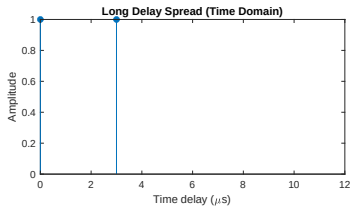
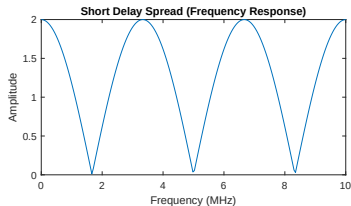
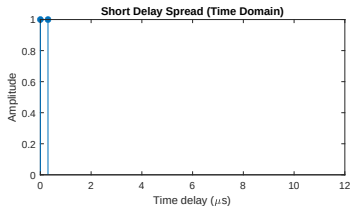
Multipath propagation (5/6)

- Two paths, each with a different amplitude.



Multipath propagation (6/6)

- Two paths, each with an amplitude equal to 1.



Orthogonality (1/10)

- Consider a set of transmitted carriers as follows:

$$\psi_n(t) = e^{j2\pi(f_0 + \frac{n}{N\Delta t})t} \text{ for } n = 0, 1, \dots, N-1 \text{ and } \Delta f = \frac{1}{N\Delta t}. \quad (2)$$

- We now show that every two carriers are orthogonal to each other. (proof is in next page)
 - In summary:

$$\int_a^b \psi_p(t) \psi_q^*(t) dt = \begin{cases} (b-a) & \text{for } p = q \\ 0 & \text{for } p \neq q \text{ and } (b-a) = N\Delta t \end{cases} \quad (3)$$

Orthogonality (2/10) — Proof

- $p = q$

$$\begin{aligned} & \int_a^b \psi_p(t) \psi_p^*(t) dt \\ &= \int_a^b e^{j2\pi(f_0 + \frac{p}{N\Delta t})t} e^{-j2\pi(f_0 + \frac{p}{N\Delta t})t} dt \\ &= \int_a^b e^0 dt \\ &= \int_a^b 1 dt \\ &= b - a \end{aligned}$$

Orthogonality (3/10) — Proof

- $p = q$, which $\psi_q(t)$ delay c

$$\begin{aligned}
 & \int_a^b \psi_p(t) \psi_p^*(t - c) dt \\
 &= \int_a^b e^{j2\pi(f_0 + \frac{p}{N\Delta t})t} e^{-j2\pi(f_0 + \frac{p}{N\Delta t})(t-c)} dt \\
 &= \int_a^b e^{j2\pi(f_0 + \frac{p}{N\Delta t})c} dt \\
 & \text{let } e^{j2\pi(f_0 + \frac{p}{N\Delta t})c} = C \text{ be a constant} \\
 &= C \int_a^b 1 dt \\
 &= C(b - a)
 \end{aligned}$$

Orthogonality (4/10) — Proof

- $p \neq q$ and $(b - a) = N\Delta t$

$$\begin{aligned} & \int_a^b \psi_p(t) \psi_q^*(t) dt \\ &= \int_a^b e^{j2\pi(f_0 + \frac{p}{N\Delta t})t} \cdot e^{-j2\pi(f_0 + \frac{q}{N\Delta t})t} dt \\ &= \int_a^b e^{j2\pi(\frac{p-q}{N\Delta t})t} dt \\ &= \frac{e^{j2\pi(p-q)\frac{b}{N\Delta t}} - e^{j2\pi(p-q)\frac{a}{N\Delta t}}}{\frac{j2\pi(p-q)}{N\Delta t}} \end{aligned}$$

Orthogonality (5/10) — Proof

- Continued from previous page:

$$= \frac{1}{j2\pi \left(\frac{p-q}{N\Delta t} \right)} \left(e^{j2\pi \left(\frac{p-q}{N\Delta t} \right) a} \cdot \left(e^{j2\pi (p-q) \frac{1}{N\Delta t} (b-a)} - 1 \right) \right)$$

$b-a$ is an observation period $= N\Delta t$

and $p-q$ is an integer so $e^{-j2\pi (p-q)} = 1$

$$= \frac{1}{j2\pi \left(\frac{p-q}{N\Delta t} \right)} \left(e^{j2\pi \left(\frac{p-q}{N\Delta t} \right) a} \cdot (1 - 1) \right)$$

$$= 0$$

Orthogonality (6/10) — Proof

- $p \neq q$ and $(b - a) = N\Delta t$, which $\psi_p(t)$ delay c and $\psi_q(t)$ delay d

$$\begin{aligned}
 & \int_a^b \psi_p(t - c) \psi_q^*(t - d) dt \\
 &= \int_a^b e^{j2\pi(f_0 + \frac{p}{N\Delta t})(t-c)} e^{-j2\pi(f_0 + \frac{q}{N\Delta t})(t-d)} dt \\
 &= \int_a^b e^{j2\pi f_0(t-c)} e^{j2\pi \frac{p}{N\Delta t}(t-c)} e^{-j2\pi f_0(t-d)} e^{-j2\pi \frac{q}{N\Delta t}(t-d)} dt \\
 &= \int_a^b \frac{e^{j2\pi \frac{p}{N\Delta t}t}}{e^{j2\pi f_0 c} e^{j2\pi \frac{p}{N\Delta t}c}} \frac{e^{-j2\pi \frac{q}{N\Delta t}t}}{e^{-j2\pi f_0 d} e^{-j2\pi \frac{q}{N\Delta t}d}} dt
 \end{aligned}$$

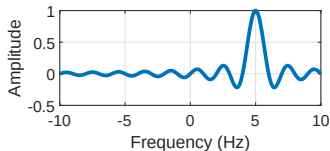
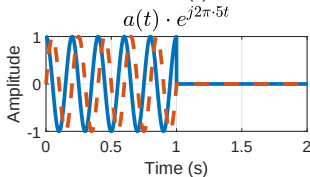
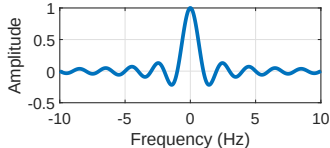
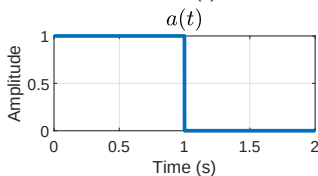
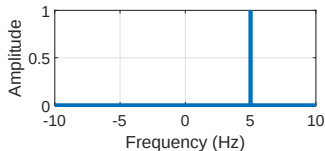
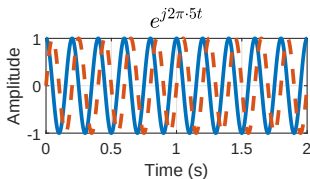
Orthogonality (7/10) — Proof

- Let $e^{-j2\pi f_0 c} e^{j2\pi f_0 d} e^{-j2\pi(\frac{p}{N\Delta t})c} e^{j2\pi(\frac{q}{N\Delta t})d} = C$ be a constant

$$\begin{aligned}
 &= C \int_a^b e^{j2\pi(\frac{p}{N\Delta t})t} e^{-j2\pi(\frac{q}{N\Delta t})t} dt \\
 &= C \int_a^b e^{j2\pi(\frac{p-q}{N\Delta t})t} dt \\
 &= C \frac{e^{j2\pi(p-q)\frac{b}{N\Delta t}} - e^{j2\pi(p-q)\frac{a}{N\Delta t}}}{\frac{j2\pi(p-q)}{N\Delta t}} \\
 &= C \frac{e^{j2\pi(p-q)\frac{b}{N\Delta t}} (1 - e^{j2\pi(p-q)\frac{1}{N\Delta t}(a-b)})}{\frac{j2\pi(p-q)}{N\Delta t}} \\
 &= 0 \text{ for } p \neq q, \text{ and } (b-a) = N\Delta t
 \end{aligned}$$

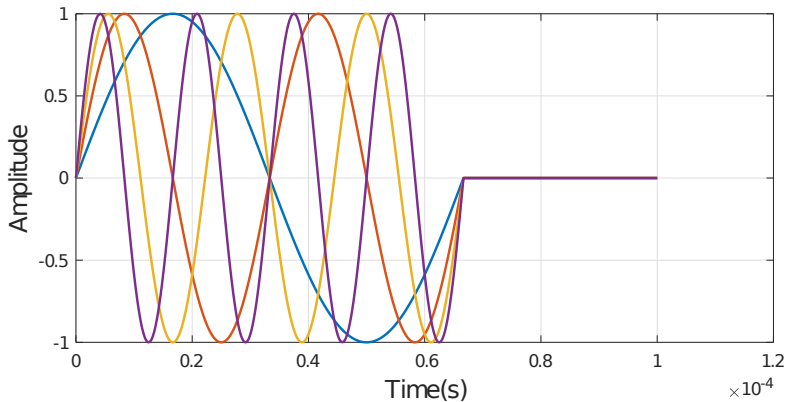
Orthogonality (8/10)

- Transmitted Signal



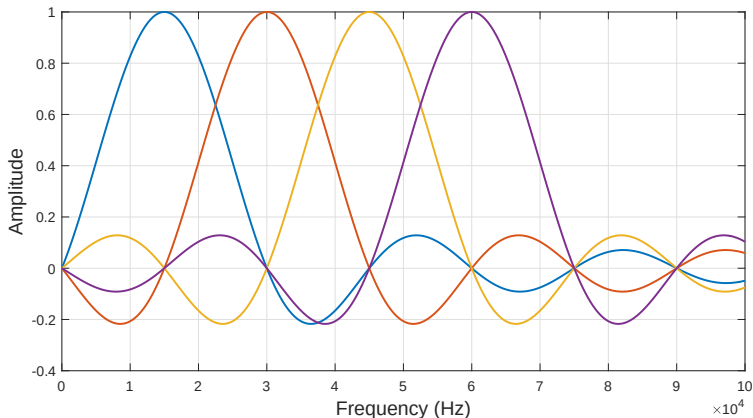
Orthogonality (9/10)

- Transmitted Signal in time domain

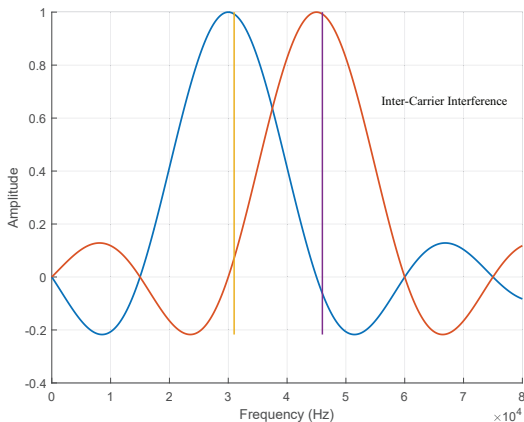


Orthogonality (10/10)

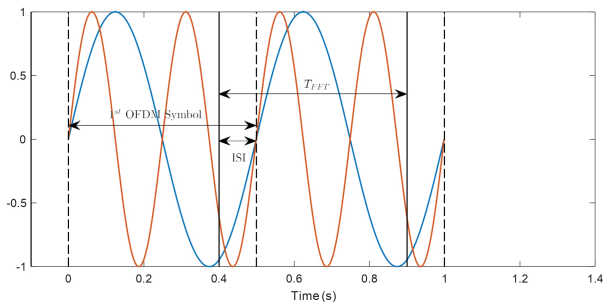
- Transmitted Signal in frequency domain



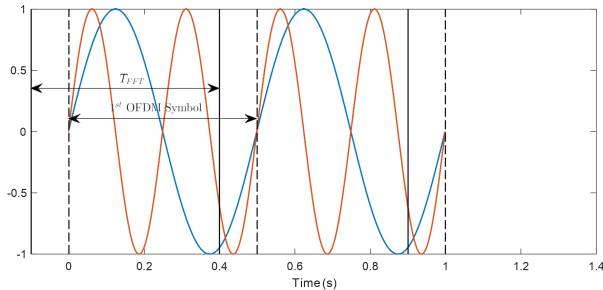
Frequency Error Result in Inter-Carrier Interference (ICI)



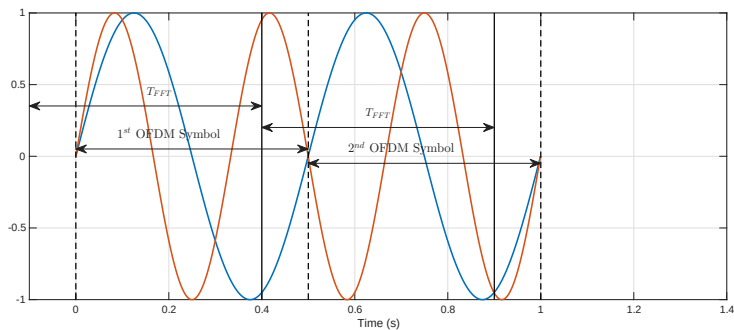
Timing Synchronization Error Results in Inter-Symbol Interference (ISI)



Timing Synchronization Error Results in Inter-Carrier Interference (ICI)

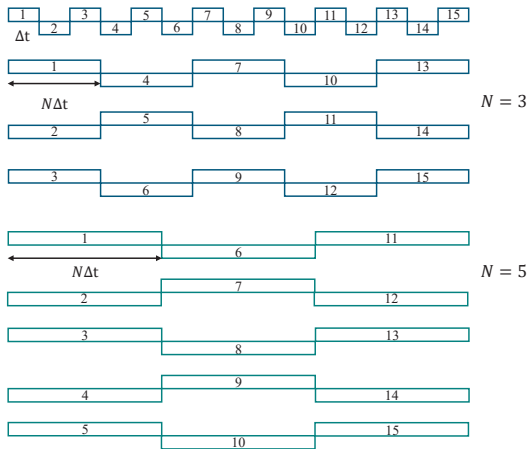


Timing Synchronization Error Results in Inter-Carrier Interference (ICI)



Mathematical Expression of OFDM Signal (1/3)

- Frequency Division Multiplexing



Mathematical Expression of OFDM Signal (2/3)

- From above, we know that $\psi_n(t)$ is the orthogonal signal set. An OFDM signal based on this orthogonal signal set can be written as:

$$x(t) = \text{Re} \left\{ \sum_{k=-\infty}^{\infty} \sum_{n=0}^{N-1} d_{k,n} \psi_n(t - kT) \right\} \quad (4)$$

where $\psi_n(t) = e^{j2\pi f_n t}$ for $n = 0, 1, \dots, N-1$

$$f_n = f_0 + \frac{n}{T}, \quad T = N\Delta t$$

$$d_{k,n} = a_{k,n} + jb_{k,n}$$

Mathematical Expression of OFDM Signal (3/3)

- T : OFDM symbol duration
- $d_{k,n}$: transmitted data on the n -th carrier of the k -th symbol.

$$x(t) = \text{Re} \left\{ \sum_{k=-\infty}^{\infty} \sum_{n=0}^{N-1} d_{k,n} \psi_n(t - kT) \right\} \quad (5)$$

$$= \sum_{k=-\infty}^{\infty} \sum_{n=0}^{N-1} \left\{ a_{k,n} \cos(2\pi f_n(t - kT)) - b_{k,n} \sin(2\pi f_n(t - kT)) \right\} \quad (6)$$

- If there is only one OFDM symbol (i.e. $k = 0$), it can be simplified as:

$$x(t) = \sum_{n=0}^{N-1} \{ a(n) \cos(2\pi f_n t) - b(n) \sin(2\pi f_n t) \} \quad (7)$$

Multi-carrier Equivalent Implementation by using IDFT (1/6)

- According to the structure of Tx, it must use N oscillators.
- That increases the hardware complexity.
- The equivalent method is using IDFT (IFFT).

Multi-carrier Equivalent Implementation by using IDFT (2/6)

- In general, each carrier can be expressed as:

$$S_c(t) = A_c(t)e^{j(2\pi f_c t + \phi_c(t))} \quad (8)$$

- We assume that there are N carriers in the OFDM signal.
- Then the total complex signal $S_s(t)$ can be represented by:

$$S_s(t) = \frac{1}{N} \sum_{n=0}^{N-1} A_n(t)e^{j(2\pi f_n t + \phi_n(t))} \quad (9)$$

- where $f_n = f_0 + n\Delta f$
- and $A_n(t)$, $\phi_n(t)$, f_n are the amplitude, phase, and carrier frequency of the n -th carrier, respectively.

Multi-carrier Equivalent Implementation by using IDFT (3/6)

- Then we sample the signal at a sampling frequency $\frac{1}{\Delta t}$, and $A_n(t)$ and $\phi_n(t)$ becomes:

$$\phi_n(t) = \phi_n \quad (10)$$

$$A_n(t) = A_n \quad (11)$$

$$S_s(k\Delta t) = \frac{1}{N} \sum_{n=0}^{N-1} A_n e^{j(2\pi(f_0+n\Delta f)k\Delta t+\phi_n)} \quad (12)$$

- Then the sampled signal can be expressed as:

$$S_s(k\Delta t) = \left(\frac{1}{N} \sum_{n=0}^{N-1} A_n e^{j\phi_n} e^{j2\pi n k \Delta f \Delta t} \right) e^{j2\pi f_0 k \Delta t} \quad (13)$$

Multi-carrier Equivalent Implementation by using IDFT (4/6)

- The inverse discrete Fourier transform (IDFT) is defined as the following:

$$f(k\Delta t) = \frac{1}{N} \sum_{n=0}^{N-1} F(n\Delta f) e^{j2\pi nk/N} \quad (14)$$

- Comparing eq.(13) and eq.(14), the condition must be satisfied in order to make eq.(13) an inverse Fourier transform relationship:

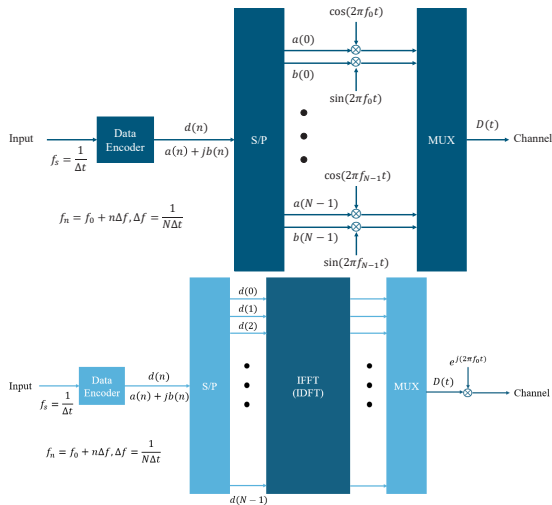
$$\Delta f = \frac{1}{N\Delta t} \quad (15)$$

Multi-carrier Equivalent Implementation by using IDFT (5/6)

- If eq.(15) is satisfied,
 - $A_n e^{j\phi_n}$ is the frequency domain signal.
 - $S_s(k\Delta t)$ is the time domain signal.
 - Δf is the sub-channel spacing.
 - $N\Delta t$ is the symbol duration in each sub-channel.
- This outcome is the same as the result obtained in the system of Figure 1. Therefore IDFT can be used to generate an OFDM transmission signal.

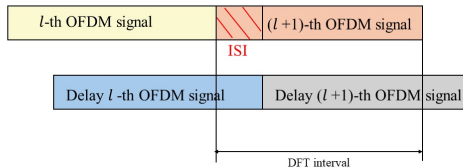
Multi-carrier Equivalent Implementation by using IDFT

(6/6)

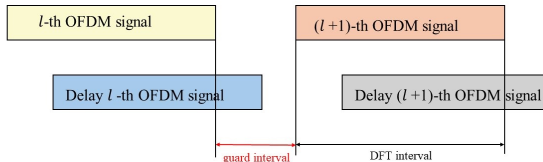


Cyclic Prefix(1/13)

- In multipath channel, delayed replicas of previous OFDM signal lead to ISI between successive OFDM signals.

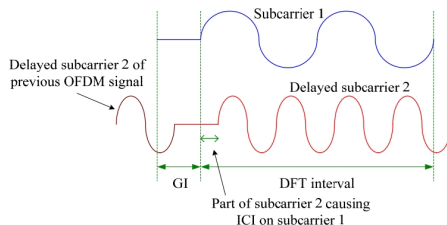


- Solution : Insert a guard interval (GI) between successive OFDM signals.



Cyclic Prefix(2/13)

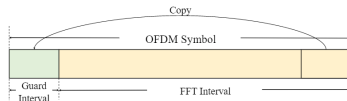
- However, adding guard interval breaks the orthogonality, leading to intercarrier interference (ICI).



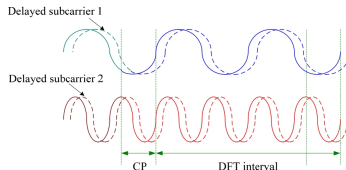
- In DFT interval, difference between two subcarriers does not maintain integer number of cycles $\rightarrow$ loss of orthogonality.
- Delayed version of subcarrier 2 causes ICI in the process of demodulating subcarrier 1.

Cyclic Prefix(3/13)

- Cyclic prefix (CP): A copy of the last part of OFDM signal is attached to the front of itself.

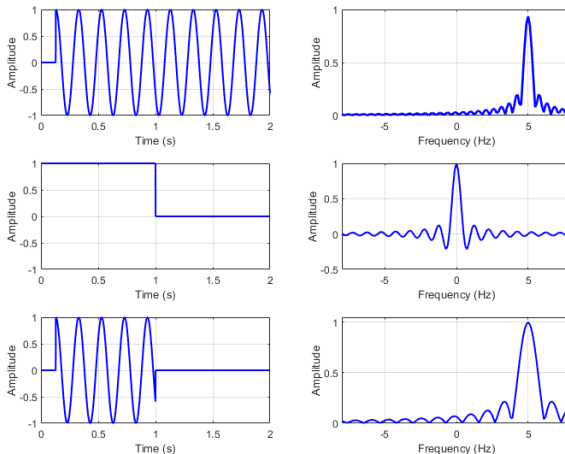


- All delayed replicas of subcarriers always have an integer number of cycles within DFT interval, which means no ICI.



Cyclic Prefix(4/13)

- Within the DFT interval, if a subcarrier does not complete an integer number of cycles, its orthogonality is lost, leading to ICI.



Cyclic Prefix(5/13)

- Original continuous subcarrier:

$$x(t) = e^{j2\pi f_0 t}, \quad -\infty < t < \infty$$

$$X(f) = \delta(f - f_0)$$

- Guard interval of duration T_{GI} :

$$a(t) = \begin{cases} 1, & T_{GI} \leq t \leq T_{GI} + T \\ 0, & \text{otherwise} \end{cases}$$

$$A(f) = T \operatorname{sinc}(fT) e^{-j2\pi f(T_{GI} + T/2)}.$$

- Subcarrier with GI:

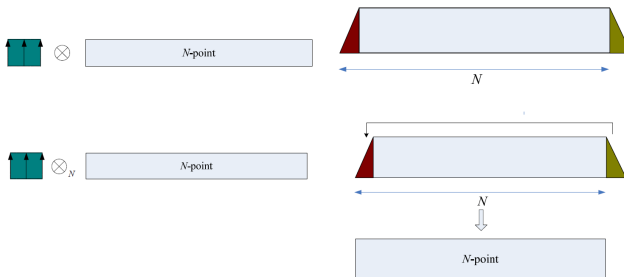
$$\hat{x}(t) = a(t) e^{j2\pi f_0 t}$$

$$\hat{X}(f) = A(f) * \delta(f - f_0) = A(f - f_0).$$

Due to GI, the spectrum becomes a shifted sinc (not a perfect delta).

Cyclic Prefix(6/13)

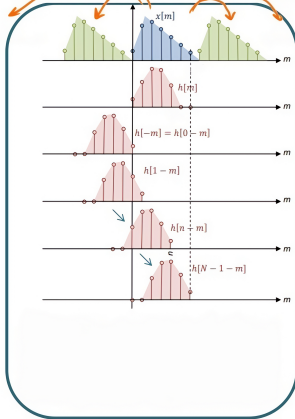
- Linear convolution vs. Circular convolution



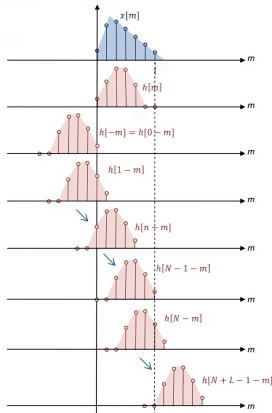
Cyclic Prefix(7/13)

- Linear convolution vs. Circular convolution

Circular Convolution



(Regular Convolution)



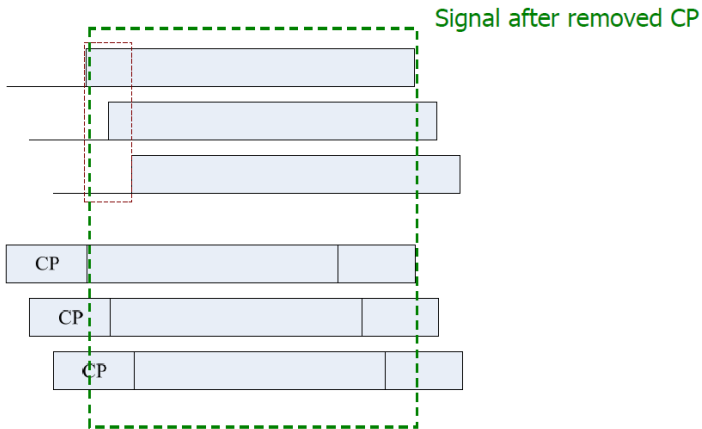
Cyclic Prefix(8/13)

- Linear convolution vs Circular convolution

Linear Convolution	Circular Convolution
$y[n] = h[n] \otimes x[n]$	$y[n] = h[n] \otimes_N x[n]$
$y[n] = \sum_{m=0}^{N-1} h[m]x[(n-m)]$	$y[n] = \sum_{m=0}^{N-1} h[m]x[((n-m))_N]$
$x[n] = [1,2,3], h[n] = [1,2,3,4,5]$ $y[n] = [\textcolor{blue}{1}, \textcolor{red}{4}, 10, 16, 22, \textcolor{blue}{22}, \textcolor{red}{15}]$	$x[n] = [1,2,3,0,0], h[n] = [1,2,3,4,5]$ $y[n] = [23, 19, 10, 16, 22]$

Cyclic Prefix(9/13)

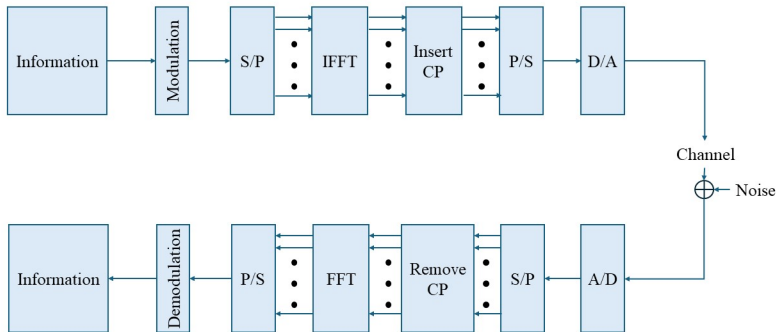
- Channel effect with cyclic prefix



Cyclic Prefix(10/13)

- Spectrum of channel $h[n]$ response with length L_h (smaller than N_g)
 $H_k = \text{FFT}\{h[n]\}$
- Received complete OFDM signal
 $\tilde{r}[n] = \tilde{D}[n] \otimes h[n], 0 \leq n \leq N + N_g + L_h - 2$
- Received useful part $r[n]$
 $r[n] = D[n] \otimes_N h[n]$
 where $\otimes_N$ is N-point circular convolution (due to CP)
- Received symbol at k-th subcarrier
 $Y_k = \text{FFT}\{r[n]\} = \text{FFT}\{D[n] \otimes_N h[n]\} = X_k H_k$
 $\Rightarrow X_k = \frac{Y_k}{H_k}$
 “Useful property for OFDM system to reduce complexity of channel equalization” .

Cyclic Prefix(11/13)



Cyclic Prefix(12/13)

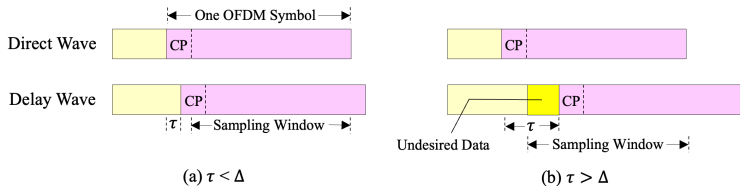
- Remove Cyclic Prefix

No Cyclic Prefix	Cyclic Prefix
$x[n] = [1,2,3,4,5], h[n] = [1,2,3]$	$x[n] = [4,5,1,2,3,4,5], h[n] = [1,2,3]$
Linear Convolution $ \begin{array}{r} 1 \ 2 \ 3 \ 4 \ 5 \\ \ 2 \ 4 \ 6 \ 8 \ 10 \\ \ 3 \ 6 \ 9 \ 12 \ 15 \\ \hline 1 \ 4 \ 10 \ 16 \ 22 \ 22 \ 15 \end{array} $	Linear Convolution $ \begin{array}{r} 4 \ 5 \ 1 \ 2 \ 3 \ 4 \ 5 \\ \ 8 \ 10 \ 2 \ 4 \ 6 \ 8 \ 10 \\ \ 12 \ 15 \ 3 \ 6 \ 9 \ 12 \ 15 \\ \hline 4 \ 13 \ 23 \ 19 \ 10 \ 16 \ 22 \ 22 \ 15 \end{array} $ Remove
Circular Convolution $ \begin{array}{r} 1 \ 2 \ 3 \ 4 \ 5 \\ 10 \ 2 \ 4 \ 6 \ 8 \\ 12 \ 15 \ 3 \ 6 \ 9 \\ \hline 23 \ 19 \ 10 \ 16 \ 22 \end{array} $	

Through CP we can get the result of Circular Convolution.

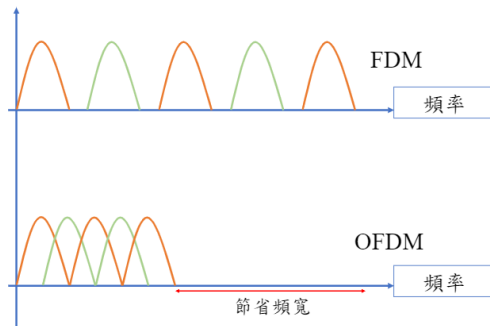
Cyclic Prefix(13/13)

- One of the most important reasons to do OFDM is the efficient way it deals with multipath delay spread.
- To eliminate inter-symbol interference (ISI) almost completely, a CP duration time (Δ) is introduced for each OFDM symbol.
(The CP duration time is chosen larger than the maximum delay spread (τ))



Bandwidth Efficiency

- In a classical parallel system, the channel is divided into N non-overlapping sub-channels to avoid inter-carrier interference (ICI).
- The diagram for bandwidth efficiency of OFDM system is shown below:



Summary

- The advantage of the FFT-based OFDM system:
 - The use of IFFT/FFT can reduce the computation complexity.
 - The orthogonality between the adjacent sub-carriers will make the use of transmission bandwidth more efficient.
 - The guard interval is used to resist the inter-symbol interference (ISI).
 - The main advantage of the OFDM transmission technique is its high performance even in frequency selective channels.
- The drawbacks of the OFDM system :
 - It is highly vulnerable to synchronization errors.
 - Peak to Average Power Ratio (PAPR) problems

$$PAPR = \frac{\text{max. power of subcarriers}}{\text{avg. power of subcarriers}}$$

Appendix

- Proof $Y_k = X_k H_k$
 - Using the definition of the DFT:

$$X_k = DFT\left\{x[n]\right\} = \sum_{n=0}^{N-1} x[n] e^{-\frac{j2\pi kn}{N}}$$

$$x[n] = IDFT\left\{X_k\right\} = \frac{1}{N} \sum_{k=0}^{N-1} X_k e^{\frac{j2\pi kn}{N}}$$

- Then, we can write the transform equation for H_k .

$$H_k = \sum_{m=0}^{N-1} h[m] e^{-\frac{j2\pi km}{N}}, \quad m = 0, 1, 2, \dots, N-1$$

Appendix

- Continued from previous page:

$$\begin{aligned}
 Y_k &= \left[\sum_{n=0}^{N-1} x[n] e^{-\frac{j2\pi kn}{N}} \right] \left[\sum_{m=0}^{N-1} h[m] e^{-\frac{j2\pi km}{N}} \right] \\
 y[n] &= \frac{1}{N} \sum_{k=0}^{N-1} \left[\sum_{l=0}^{N-1} x[l] e^{-\frac{j2\pi kl}{N}} \right] \left[\sum_{m=0}^{N-1} h[m] e^{-\frac{j2\pi km}{N}} \right] e^{\frac{j2\pi kn}{N}} \\
 &= \frac{1}{N} \sum_{l=0}^{N-1} x[l] \sum_{m=0}^{N-1} h[m] \left[\sum_{k=0}^{N-1} e^{\frac{j2\pi k(n-l-m)}{N}} \right]
 \end{aligned}$$

Appendix

- Now, we see that the item in brackets can be evaluated as

$$\sum_{k=0}^{N-1} e^{\frac{j2\pi k(n-l-m)}{N}} = \begin{cases} N, & n-l-m = KN, \text{ for some integer } K \\ 0, & \text{otherwise} \end{cases}$$

and, therefore

$$y[n] = \frac{1}{N} \sum_{l=0}^{N-1} x[l] \sum_{m=0}^{N-1} h[m] N \delta[n-l-m-KN]$$

$$(n-l-m) \bmod N \equiv 0 \bmod N$$

Because the impulse function is zero expect when

$$m \bmod N \equiv (n-l) \bmod N$$

, we can rewrite the equation as

$$y[n] = \sum_{l=0}^{N-1} x[l] h[n-l]_{\bmod N}$$

Appendix

- Circular Convolution in Time Domain

$$y[n] = \sum_{m=0}^{N-1} x[m] h[(n - m) \bmod N], \quad n = 0, \dots, N - 1$$

Take the DFT of $y[n]$:

$$\begin{aligned} Y_k &= \sum_{n=0}^{N-1} y[n] e^{-j \frac{2\pi}{N} kn} \\ &= \sum_{n=0}^{N-1} \sum_{m=0}^{N-1} x[m] h[(n - m) \bmod N] e^{-j \frac{2\pi}{N} kn}. \end{aligned}$$

Appendix

- Continued from previous page:

Let $r = (n - m) \bmod N$

$$\begin{aligned} Y_k &= \sum_{m=0}^{N-1} x[m] \sum_{r=0}^{N-1} h[r] e^{-j \frac{2\pi}{N} k(r+m)} \\ &= \left(\sum_{m=0}^{N-1} x[m] e^{-j \frac{2\pi}{N} km} \right) \left(\sum_{r=0}^{N-1} h[r] e^{-j \frac{2\pi}{N} kr} \right) \\ &= X_k H_k. \end{aligned}$$

$$\boxed{Y_k = X_k H_k}$$

Appendix

- Circular convolution:

$$y[n] = x[n] \underset{N}{\otimes} h[n] = \sum_{l=0}^{N-1} x[l]h[n-l]_{\text{mod } N}$$

As a result, we can get the DFT pair:

$$y[n] = x[n] \underset{N}{\otimes} h[n] \leftrightarrow Y_k = X_k H_k$$

Appendix

- Integrate the Exponential Function

To integrate this, we use the formula for the integral of an exponential function:

$$\int e^{j\omega t} dt = \frac{e^{j\omega t}}{j\omega}$$

Here, $\omega = 2\pi \frac{p-q}{N\Delta t}$. Applying this to our integral:

$$\int_a^b e^{j2\pi(\frac{p-q}{N\Delta t})t} dt = \left[\frac{e^{j2\pi(\frac{p-q}{N\Delta t})t}}{j2\pi(\frac{p-q}{N\Delta t})} \right]_a^b$$

Appendix

- Evaluate the Integral from a to b
Substituting the limits a and b into the expression:

$$\frac{1}{j2\pi\left(\frac{p-q}{N\Delta t}\right)} \left(e^{j2\pi\left(\frac{p-q}{N\Delta t}\right)b} - e^{j2\pi\left(\frac{p-q}{N\Delta t}\right)a} \right)$$

Appendix

- Given that $b = a + N\Delta t$, substitute this into the expression

$$e^{j2\pi\left(\frac{p-q}{N\Delta t}\right)b} = e^{j2\pi\left(\frac{p-q}{N\Delta t}\right)(a+N\Delta t)}$$

Utilizing the property of exponents $e^{j(\alpha+\beta)} = e^{j\alpha}e^{j\beta}$, we can rewrite the expression as:

$$e^{j2\pi\left(\frac{p-q}{N\Delta t}\right)b} = e^{j2\pi\left(\frac{p-q}{N\Delta t}\right)a} \cdot e^{j2\pi(p-q)}$$

Appendix

- Since $p - q$ is an integer, $e^{j2\pi(p-q)} = 1$. This is due to the periodic nature of the complex exponential function, where the result returns to 1 every time the angle is a multiple of 2π

$$e^{j2\pi\left(\frac{p-q}{N\Delta t}\right)b} = e^{j2\pi\left(\frac{p-q}{N\Delta t}\right)a}$$

substitute these results back into the integral expression

$$\frac{1}{j2\pi\left(\frac{p-q}{N\Delta t}\right)} \left(e^{j2\pi\left(\frac{p-q}{N\Delta t}\right)a} \cdot e^{j2\pi(p-q)} - e^{j2\pi\left(\frac{p-q}{N\Delta t}\right)a} \right)$$

Rearranging terms:

$$\frac{1}{j2\pi\left(\frac{p-q}{N\Delta t}\right)} \left(e^{j2\pi\left(\frac{p-q}{N\Delta t}\right)a} \cdot (1 - 1) \right)$$

Appendix

- Result

Since $e^{j2\pi(p-q)} = 1$, we have $1 - 1 = 0$. Thus, the entire expression ultimately simplifies to:

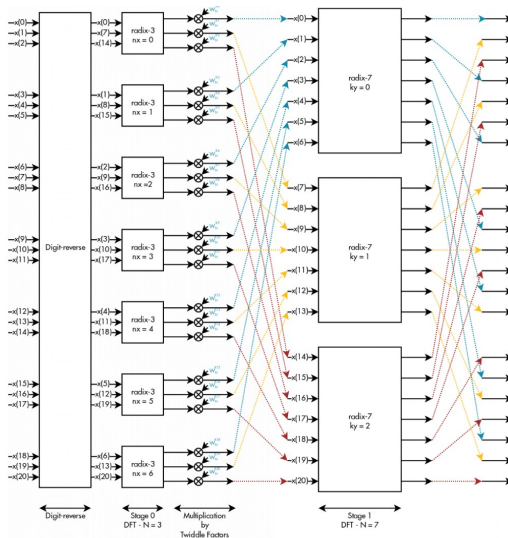
$$\frac{1}{j2\pi \left(\frac{p-q}{N\Delta t} \right)} \cdot 0 = 0$$

Appendix

- The Mixed-Radix Cooley-Tukey algorithm
 - Ideal for lengths that are not powers of two, it efficiently handles composite lengths.
 - Breaking down a DFT of size $N = N_1 \times N_2$ into smaller DFTs of sizes N_1 and N_2 .
 - The algorithm reduces the time complexity from $O(N^2)$ to $O(N_1 \times N_2 \times (N_1 + N_2)) = O(N \times (N_1 + N_2))$.

Appendix

- Mixed-Radix Cooley-Tukey Algorithm: $N = 21, N_1 = 7, N_2 = 3$



Appendix

- The Mixed-Radix Cooley-Tukey algorithm is defined as the DFT:

$$X_k = \sum_{n=0}^{N-1} x_n e^{-\frac{2\pi i}{N}nk} \quad \text{for } k = 0, \dots, N-1$$

Given that $N = N_1 \cdot N_2$

Substituting $n = n_1 + n_2 N_1$ and $k = k_1 N_2 + k_2$, we have:

$$\begin{aligned} e^{-\frac{2\pi i}{N}nk} &= e^{-\frac{2\pi i}{N}(n_1+n_2N_1)(k_1N_2+k_2)} \\ &= e^{-\frac{2\pi i}{N}n_1k_1N_2} e^{-\frac{2\pi i}{N}n_1k_2} e^{-\frac{2\pi i}{N}n_2N_1k_1N_2} e^{-\frac{2\pi i}{N}n_2N_1k_2} \\ &= e^{-\frac{2\pi i}{N_1}n_1k_1} e^{-\frac{2\pi i}{N}n_1k_2} e^{-\frac{2\pi i}{N_2}n_2k_2} \end{aligned}$$

- The Mixed-Radix Cooley-Tukey Algorithm:

$$X_{k_1N_2+k_2} = \sum_{n_1=0}^{N_1-1} \left(\sum_{n_2=0}^{N_2-1} \left(x_{n_1+n_2N_1} e^{-\frac{2\pi i}{N_2}n_2k_2} \right) e^{-\frac{2\pi i}{N}n_1k_2} e^{-\frac{2\pi i}{N_1}n_1k_1} \right)$$

Appendix

- The Example of Cooley-Tukey Algorithm:

Given that $N = 15$, $N_1 = 3$, $N_2 = 5$

Substituting $n = n_1 + 3n_2$ and $k = 5k_1 + k_2$, we have:

$$\begin{aligned}
 e^{-\frac{2\pi i}{N}nk} &= e^{-\frac{2\pi i}{15}(n_1+3n_2)(5k_1+k_2)} \\
 &= e^{-\frac{2\pi i}{15}5n_1k_1} e^{-\frac{2\pi i}{15}n_1k_2} e^{-\frac{2\pi i}{15}15n_2k_1} e^{-\frac{2\pi i}{15}3n_2k_2} \\
 &= e^{-\frac{2\pi i}{3}n_1k_1} e^{-\frac{2\pi i}{15}n_1k_2} e^{-\frac{2\pi i}{5}n_2k_2}
 \end{aligned}$$

- The Mixed-Radix Cooley-Tukey Algorithm:

$$X_{5k_1+k_2} = \sum_{n_1=0}^{3-1} \left(\sum_{n_2=0}^{5-1} \left(x_{n_1+3n_2} e^{-\frac{2\pi i}{5}n_2k_2} \right) e^{-\frac{2\pi i}{15}n_1k_2} e^{-\frac{2\pi i}{3}n_1k_1} \right)$$